

# Redistricting Analysis: Rhode Island's 2018 and 2020 Elections

## Author

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## Github Repo

[https://github.com/JinglinZhou99/Final\\_Project\\_MAUP\\_RI](https://github.com/JinglinZhou99/Final_Project_MAUP_RI)

## Introduction

Redistricting plays a vital role in shaping political representation and ensuring equitable voting power across geographic regions. This report examines the redistricting process in Rhode Island by analyzing the state's enacted 2022 State House district map in comparison to thousands of alternative districting plans generated using computational methods. By leveraging the GerryChain library and applying ensemble and optimization techniques, we explore how different redistricting outcomes affect partisan balance, compactness, and minority representation—especially focusing on Latino-majority districts. Using data from the 2018 U.S. Senate and 2020 Presidential elections, this analysis aims to assess the fairness of the enacted plan and demonstrate the broader implications of redistricting on democratic representation.

## State Background: Rhode Island

### 1. Political Inclinations

Rhode Island is broadly considered a strongly Democratic state. Since the Great Depression era, the Democratic Party has largely dominated state-level politics. In presidential elections, the state has consistently voted for the Democratic candidate since 1928, with only four exceptions (1952, 1956, 1972, 1984).

Currently, Rhode Island operates under a Democratic trifecta, meaning the Democratic Party controls the office of the Governor and both chambers of the state legislature (the General Assembly). This dominance is particularly pronounced in the **Rhode Island House of Representatives**, where Democrats have held the majority continuously since 1940. As of the 2024 elections, the House consists of 64 Democrats, 10 Republicans, and 1 Independent, maintaining a significant Democratic supermajority. While the state remains a Democratic stronghold, recent analysis of the 2024 election results indicated some Republican gains in voter share across various municipalities compared to previous cycles.

### 2. Population Growth

As the smallest state in the United States by land area, Rhode Island has the second-highest population density. Its overall population growth has been relatively modest, especially compared to rapidly growing states. The state's population hovered around 1.1 million residents as of 2023-2024.

Historical data shows periods of growth followed by slower phases. From 1990 to 2020, Rhode Island's population grew, but at a much slower pace than the national average. Recent trends indicate slight increases year-over-year, primarily driven by growth in Providence County, the state's most populous county. Some other areas, particularly Newport County, have experienced slight population declines in recent census estimates. The state also reflects national trends of an aging population.

### 3. Population Racial Demographics

Rhode Island's population is predominantly White (Non-Hispanic), comprising roughly 69-72% of residents according to recent estimates (U.S. Census Bureau, Data USA). However, the state has diverse and growing minority communities. Significant groups include:

- **Hispanic or Latinx:** Approximately 16-18%
- **Black or African American:** Approximately 5-9%
- **Asian:** Approximately 3-4%
- **Two or More Races:** Approximately 3-10% (varying estimates based on methodology)
- **Native American:** Less than 1%

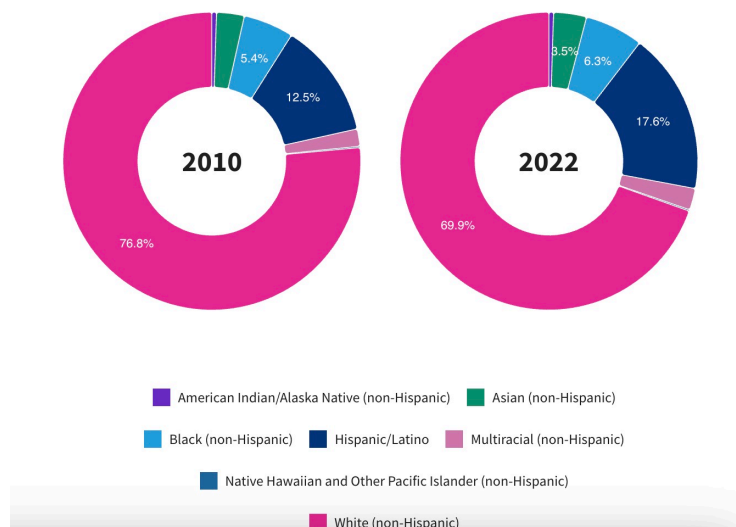


Figure 1: Sample of Racial Makeup of RI (USAFacts, 2025)

Much of Rhode Island's population growth over the past few decades can be attributed to increases in communities of color, particularly the Latinx population. This growth has been notable even as the state's overall birth rate has decreased. The largest communities of color are concentrated in urban areas such as Providence, Pawtucket, Central Falls.

#### **4. District History & Redistricting Process (State House Focus)**

In Rhode Island, the power to draw and enact district boundaries for both the U.S. Congress and the state legislature (State Senate and **State House of Representatives**) rests primarily with the **Rhode Island General Assembly** (the state legislature).

- **Process Authority:** Rhode Island uses a legislature-dominant system. While an 18-member advisory "Special Commission on Reapportionment" was established in 2011 (and re-established for the 2020 cycle) to study population shifts, hold hearings, and make recommendations, the General Assembly is not bound by these recommendations and retains the ultimate authority to pass redistricting plans as legislation.
- **Gubernatorial Role:** Enacted plans are subject to the Governor's veto, though the General Assembly can override a veto with a three-fifths vote in both the House and Senate. Given the current Democratic supermajorities, overrides are generally feasible if needed.
- **Criteria:** The Rhode Island Constitution requires that districts shall be "substantially equal in population" and "as compact in territory as possible." Federal requirements like the Voting Rights Act also apply, prohibiting discrimination based on race or ethnicity.
- **Timeline:** Redistricting occurs every ten years following the U.S. Census. The most recent redistricting cycle followed the 2020 Census, with new maps enacted in early 2022 for use starting with that year's elections.
- **Recent History (2020 Cycle):** The Special Commission on Reapportionment held public hearings and released draft maps in late 2021 and early 2022. The General Assembly then took up the process, passing final legislative plans (including for the 75 State House districts) on February 15, 2022. Governor Dan McKee signed the legislation the following day.

#### **5. Enacted State House District Map (2022)**

The current map outlining the 75 districts for the Rhode Island House of Representatives was adopted on February 16, 2022. This map reflects population changes recorded in the 2020 U.S. Census and aims to meet state and federal requirements for equal population and compactness, while also considering communities of interest.

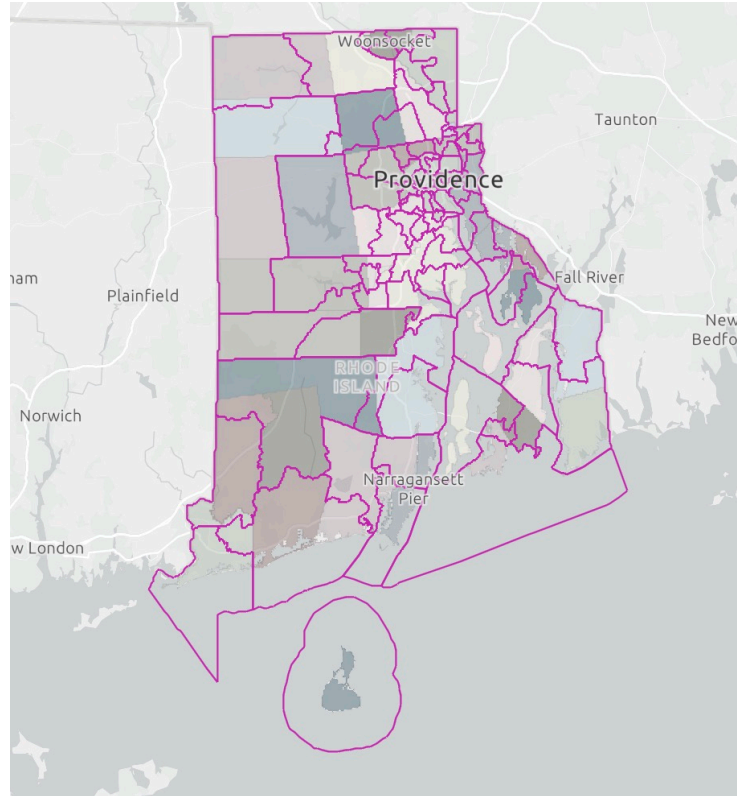


Figure 2: Rhode Island State House Map (<https://www.rilegislature.gov/>)

## Data Collection and Preparation for Rhode Island Analysis

### 1. Introduction

This section details the data collection and preparation steps performed in the Jupyter Notebook `MAUP_RI.ipynb`. The primary goal of this section is to gather, clean, and align various spatial and tabular datasets for Rhode Island, preparing them for subsequent spatial analysis, particularly concerning redistricting and the Modifiable Areal Unit Problem (MAUP). The process involves loading different data layers, standardizing their projections, repairing geometric inconsistencies, and aggregating/disaggregating data across different geographic units (blocks, precincts, house districts).

### 2. Data Sources

We utilize several key datasets for Rhode Island, obtained from the Redistricting Data Hub:

- **2020 Census Population Data (PL 94-171):** Loaded at the Census Block level. This includes total population and Hispanic/Race data.
  - **Source:** `ri_pl2020_b/ri_pl2020_p2_b.shp`

- *Source:* `ri_pl2020_b/ri_pl2020_p4_b.shp`
- *URL:*  
<https://redistrictingdatahub.org/dataset/rhode-island-block-pl-94171-2020-by-table/>
- **2022 State House Districts:** The approved State House legislative district boundaries.
  - *Source:* `ri_sldl_adopted_2022/RI_House_D_SubA.shp`
  - *URL:*  
<https://redistrictingdatahub.org/dataset/2022-rhode-island-house-of-representatives-districts-approved-plan/>
- **2020 Election Data (VEST):** Precinct-level election results for the 2020 general election.
  - *Source:* `ri_vest_20/ri_vest_20.shp`
  - *URL:*  
<https://redistrictingdatahub.org/dataset/vest-2020-rhode-island-precinct-and-election-results/>
- **2018 Election Data (VEST):** Precinct-level election results for the 2018 general election.
  - *Source:* `ri_vest_18/ri_vest_18.shp`
  - *URL:*  
<https://redistrictingdatahub.org/dataset/vest-2018-rhode-island-precinct-and-election-results/>

### 3. Libraries Used

The preparation process relies heavily on the following Python libraries:

- **pandas:** For data manipulation and analysis.
- **geopandas:** For working with geospatial data (reading shapefiles, managing geometries, spatial operations).
- **maup:** A specialized library for redistricting data processing, including geometry cleaning, data aggregation/disaggregation between different geographic levels, and diagnosing spatial issues.
- **time, os, warnings:** Standard Python libraries for timing operations, interacting with the operating system (file paths), and managing warnings.

### 4. Key Preparation Steps

The notebook executes the following key sequence of steps to prepare the data:

1. **Standardize CRS:** All input GeoDataFrames (population blocks, VAP blocks, precincts, house districts) are converted to a common projected Coordinate

Reference System (CRS) using `estimate_utm_crs()` and `to_crs()`. This ensures spatial consistency crucial for accurate overlays and assignments.

2. **Diagnose Geometries:** The `maup.doctor()` function is used to check the geometric validity of each layer. This step identifies issues like self-intersections, overlaps, or gaps that can interfere with spatial analysis. The State House district layer (`house_df`) was found to have initial geometric problems (overlaps and holes).
3. **Repair Geometries:** The `maup.smart_repair()` function is applied to the State House district layer (`house_df`) to automatically fix detected overlaps and fill gaps, creating a topologically valid layer (`repaired_house_df`). A second repair step is performed to convert very small shared boundaries (rook adjacencies below a 30-unit threshold) into single points (queen adjacencies), further improving geometric integrity for assignment purposes. ***Note: Nesting precincts within counties was avoided in the workflow, as preliminary tests indicated this could introduce persistent holes in the `house_df` layer even after repair attempts.***
4. **Assign Blocks to Precincts:** `maup.assign()` determines the spatial relationship between 2020 Census blocks and 2020 VEST precincts. It creates a mapping (Series) indicating which precinct each block primarily belongs to. Separate assignments are created for total population and VAP blocks.
5. **Aggregate Population/VAP Data:** Block-level 2020 Census data (total population and VAP by race/ethnicity) is aggregated to the 2020 precinct level using the block-to-precinct assignments and `groupby().sum()`. These aggregated demographic totals are added as new columns to the 2020 precinct GeoDataFrame (`vest20_df`). Sums are checked to ensure data integrity.
6. **Prorate 2018 Election Data:** Since 2018 precincts may not align perfectly with 2020 blocks, 2018 election results are disaggregated from the 2018 precinct level down to the 2020 block level. This is done using `maup.prorate()`, weighted by the Voting Age Population (VAP) calculated for the block-to-2018-precinct intersections.
7. **Aggregate 2018 Data to 2020 Precincts:** The prorated 2018 election data (now at the 2020 block level) is aggregated up to the 2020 precinct level using the block-to-2020-precinct assignment (`blocks_to_precincts_assignment`) and `groupby().sum()`. This adds the estimated 2018 results to the primary `vest20_df`. Totals are compared to the original 2018 data as a check.
8. **Assign Precincts to House Districts:** `maup.assign()` is used again to determine which *repaired* 2022 State House district each 2020 precinct belongs to based on spatial intersection.
9. **Add District ID to Precincts:** A new column (HD) is added to the `vest20_df`, storing the corresponding State House District ID obtained from the assignment

step. The code ensures the correct district label (e.g., '1', '75') is used, not just the index.

10. **Save Prepared Data:** The final, processed `vest20_df` GeoDataFrame, now containing 2020 precinct geometries along with aggregated 2020 demographics, 2020 election results, aggregated 2018 election results, and the assigned 2022 House District ID, is saved as a shapefile (`./RI/RI.shp`) in a newly created `./RI` directory.

## GerryChain Analysis of Rhode Island's Redistricting using 2018 Senate and 2020 Presidential Election Data

### 1. Introduction

This section explains the analyses performed in the Jupyter notebooks `gerrychainRI_2018.ipynb` and `gerrychainRI_2020.ipynb`. Both notebooks utilize the GerryChain library to evaluate Rhode Island's enacted 2022 State House districting plan. The core methodology involves generating an ensemble of 10,000 alternative, randomly generated districting plans and comparing the enacted plan's characteristics against this ensemble.

The key difference between the notebooks lies in the election data used for partisan analysis:

- `gerrychainRI_2018.ipynb` uses the 2018 US Senate election results.
- `gerrychainRI_2020.ipynb` uses the 2020 Presidential election results.

Both analyses focus on metrics such as population balance, compactness (measured by cut edges), potential for Latino majority districts, and partisan outcomes (Democratic-won districts and the Efficiency Gap). The following sections outline the methodology, explain key concepts, and summarize the results.

### 2. Key Concepts Explained

- **ReCom (Recombination Algorithm):** This is a Markov Chain Monte Carlo (MCMC) method used to generate alternative districting plans. It works by randomly selecting two adjacent districts, merging them into a single unit, and then re-splitting this unit into two new districts while attempting to maintain population balance (within a defined tolerance) and contiguity. This process is repeated many times to create an ensemble of diverse, legally valid plans.
- **Cut Edges:** This metric quantifies the compactness of a districting plan. In a graph where precincts (or other geographic units) are nodes and shared boundaries are edges, "cut edges" are the edges that cross district lines. A lower



number of cut edges generally indicates more compact districts with smoother, more regular boundaries, as districts cut across fewer underlying unit boundaries.

- **Efficiency Gap (EG):** This is a measure of partisan fairness that quantifies "wasted votes" for each party in an election. Wasted votes include all votes for a losing candidate and any votes for a winning candidate beyond the 50%+1 needed to win. The Efficiency Gap is calculated as the difference between the two parties' total wasted votes, divided by the total number of votes cast.
  - A **negative EG** typically indicates that the plan favors the Democratic party (i.e., Republicans have more wasted votes, or Democrats have fewer).
  - A **positive EG** typically indicates that the plan favors the Republican party.
  - An EG **close to zero** is generally considered more partisanly "fair" or "efficient." An EG beyond +/-8% is often cited as potentially indicative of a partisan gerrymander.

### 3. Methodology

Both notebooks follow a similar analytical procedure:

#### 1. Data Loading and Graph Construction:

- The Rhode Island precinct-level shapefile (`RI.shp`), containing geometries, demographic data (Total Population `TOTPOP`, Hispanic Population `HISP`), and election results for both 2018 and 2020, is loaded using `geopandas`.
- This shapefile is converted into a spatial graph where precincts are nodes and shared boundaries are edges. A warning about a disconnected node (island 91) is noted in both notebooks, but the analysis proceeds by ignoring errors.

#### 2. Initial Partition (Enacted Plan):

- The enacted 2022 State House districting plan, defined by the `HD` (House District) assignment column in the shapefile, is used as the starting point (`initial_partition`).

#### 3. Updaters (Metrics to Track):

- Common updaters include: Total Population (`population`), Cut Edges (`cut_edges`), and Hispanic Population (`latino_population`).
- Election-specific updaters:
  - **2018 Notebook:** `sen_2018` using `G18USSDWHI` (Democratic US Senate votes) and `G18USSRFLA` (Republican US Senate votes).
  - **2020 Notebook:** `pres_2020` using `G20PRED` (Democratic Presidential votes) and `G20PRER` (Republican Presidential votes).

#### 4. Population Analysis:



- The ideal population per district is calculated as total state population / 75 (approx. 14,631.72).
- **A population deviation tolerance is set at 30%.** This relatively large tolerance is necessary because the most populous district contains 18,642 residents, significantly above the average of 14,631.72.

#### 5. Markov Chain Simulation (Ensemble Generation):

- The **ReCom** proposal method is used.
- A population constraint (`constraints.within_percent_of_ideal_population`) ensures generated plans adhere to the specified population deviation.
- A **short chain** of 10,000 steps is run, accepting every valid proposed plan (`accept=accept.always_accept`), to generate an ensemble for analyzing cut edges, majority-Latino districts, and Democratic-won districts.
- A **long chain** of 40,000 steps is run specifically to observe the convergence of the Efficiency Gap metric.

#### 6. Metrics Tracked per Plan in the Ensemble:

- Number of cut edges.
- Number of majority-Latino districts (where Latino population > 50% of the district's total population).
- Number of Democratic-won districts (where Democratic vote share > 50% for the respective election).
- Efficiency Gap.
- Sorted Democratic vote percentages for each district (used for box plots).

#### 7. Output:

- Results are stored in a `pandas.DataFrame`.
- Averages for the tracked metrics are computed.
- Visualizations include histograms for cut edges, majority-Latino districts, Democratic-won districts, Efficiency Gap, box plots for Democratic vote shares, and convergence plots for the Efficiency Gap.

### 4. Results and Explanation of Visualizations

#### Initial Enacted Plan Metrics:

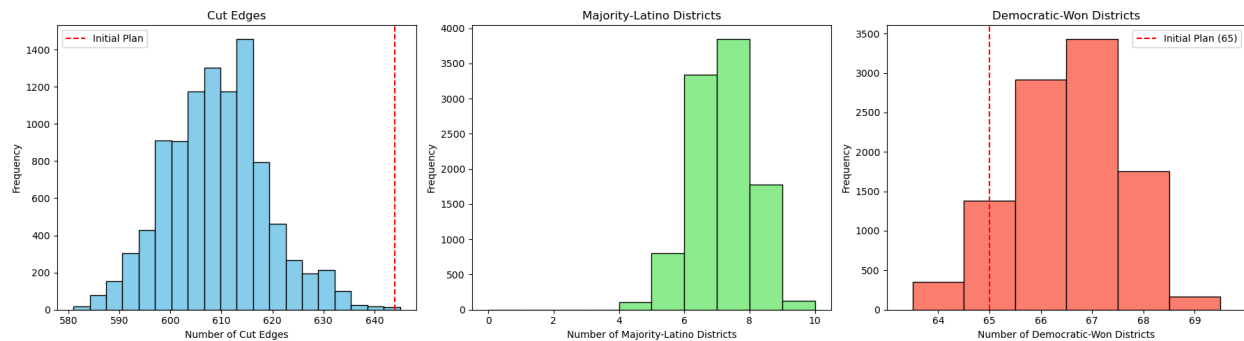
- **Population:** All 75 districts in the enacted plan are population-balanced within the specified tolerance.
- **Partisan Outcomes (Enacted Plan):**
  - **2018 Senate Election:** 65 out of 75 districts were Democratic-won.
  - **2020 Presidential Election:** 63 out of 75 districts were Democratic-won.
- **Efficiency Gap (Enacted Plan):** The notebooks calculate this for the initial plan, and it is marked on the EG histograms.

### Ensemble Averages (from 10,000-step chains):

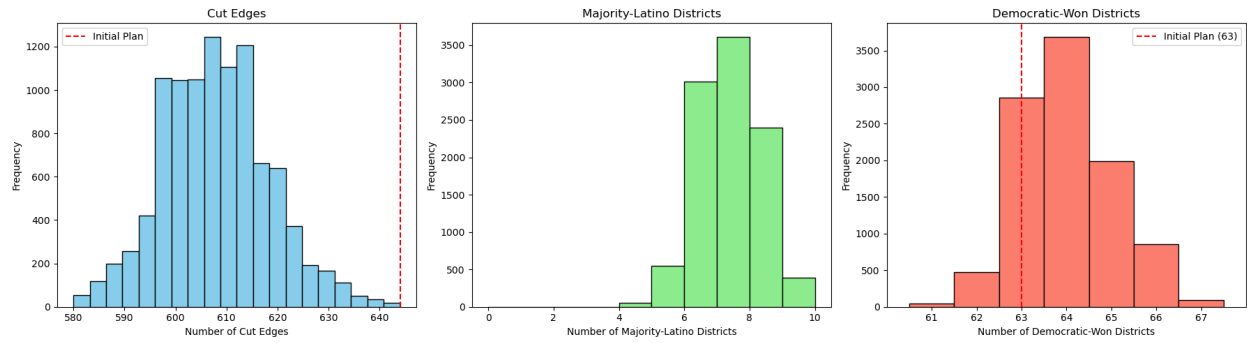
| Metric                    | 2018 Senate Data<br>(Ensemble Avg.) | 2020 Presidential Data<br>(Ensemble Avg.) |
|---------------------------|-------------------------------------|-------------------------------------------|
| Cut Edges                 | 609                                 | 608                                       |
| Majority-Latino Districts | 7                                   | 7                                         |
| Democratic-Won Districts  | 67                                  | 64                                        |
| Efficiency Gap            | -0.1291 (-12.91%)                   | -0.1164 (-11.64%)                         |

(Note: Negative EGs indicate a Republican advantage in the ensemble plans.)

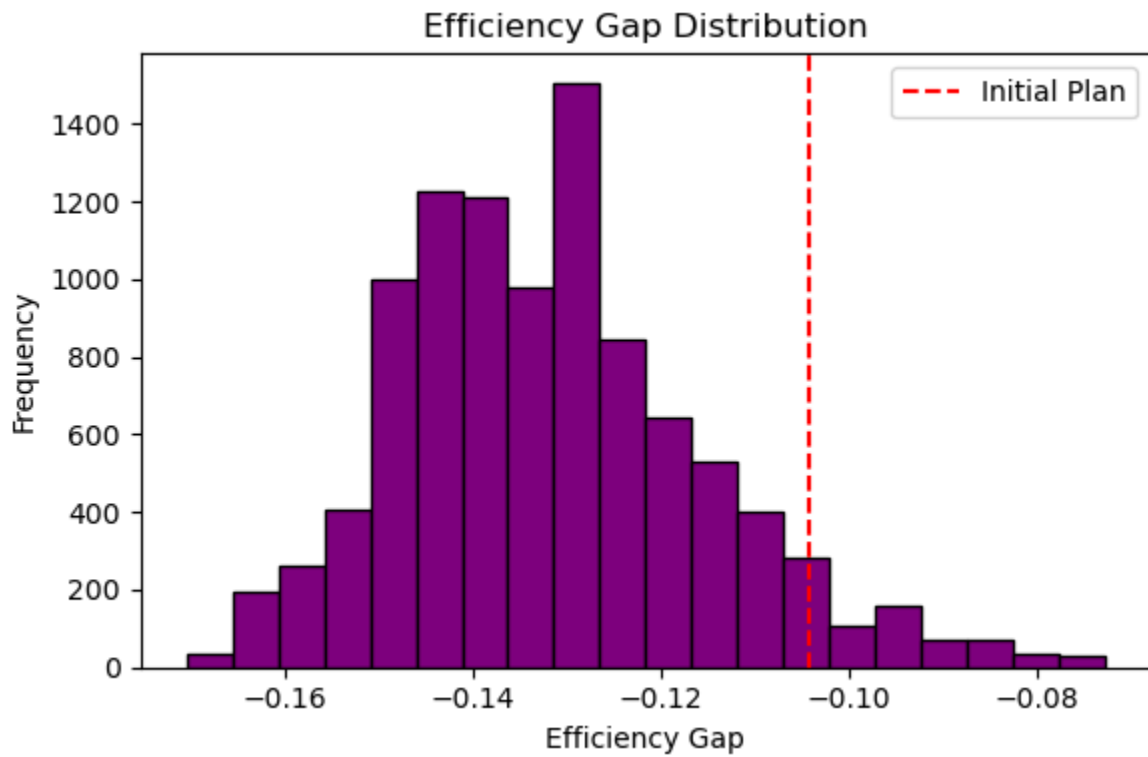
### Histograms:



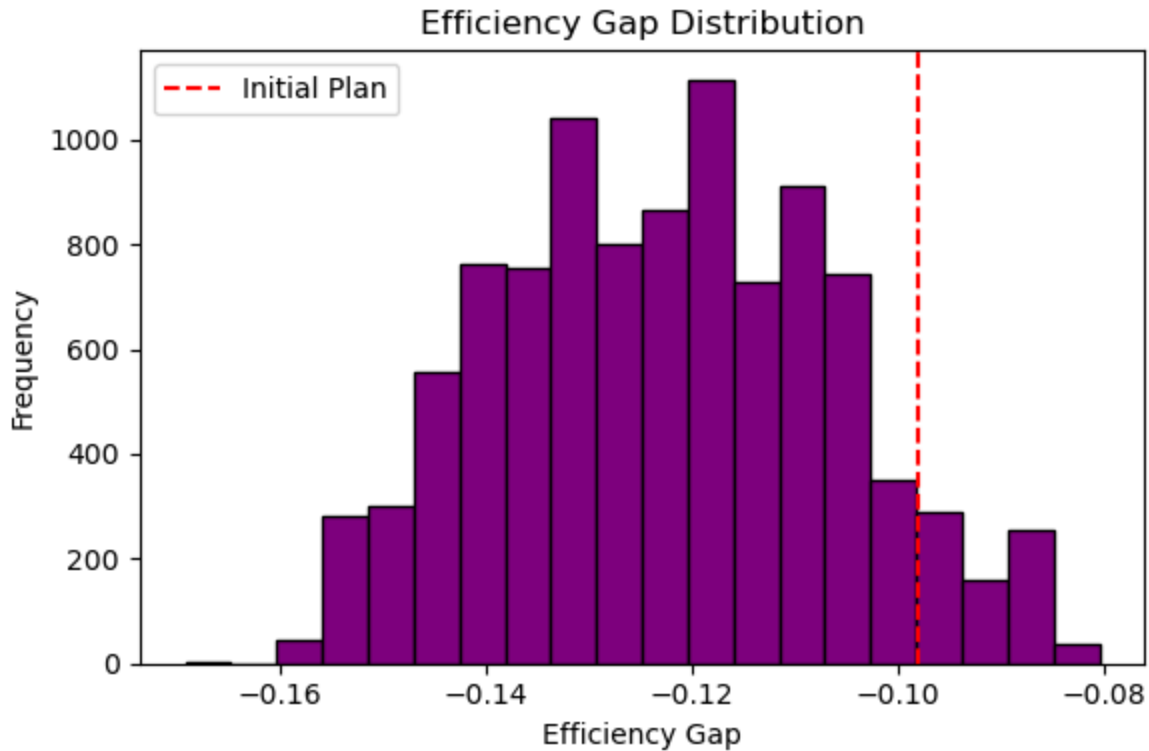
**Figure 3: 2018 Senate Election Histograms**



**Figure 4: 2018 Senate Election Histograms**



**Figure 5: 2018 Efficiency Gap Distribution**



**Figure 6: 2020 Efficiency Gap Distribution**

The histograms illustrate the distribution of these metrics across the 10,000 generated plans. A red dashed line on some plots indicates the value for the initial (enacted) plan.

### 1. Cut Edges Histogram:

- **What It Shows:** Displays the frequency of different cut edge counts. Fewer cut edges suggest more compact districts.
- **Interpretation (2018 Data):** The cut edges of the initial plan are significantly higher than most simulation plans, which means that its compactness may be poor.
- **Interpretation (2020 Data):** Likewise, the cut edges of the initial solution are high, indicating a high boundary complexity, which is not as compact as most of the generated solutions.
- **Why It Matters:** This helps assess if the enacted plan's district shapes are reasonable compared to typical alternatives generated under neutral criteria. A high cut edge count for the enacted plan relative to the ensemble suggests it is more sprawling or less compact.

### 2. Majority-Latino Districts Histogram:

- **What It Shows:** Shows the frequency of plans with different numbers of districts where the Latino population exceeds 50%.

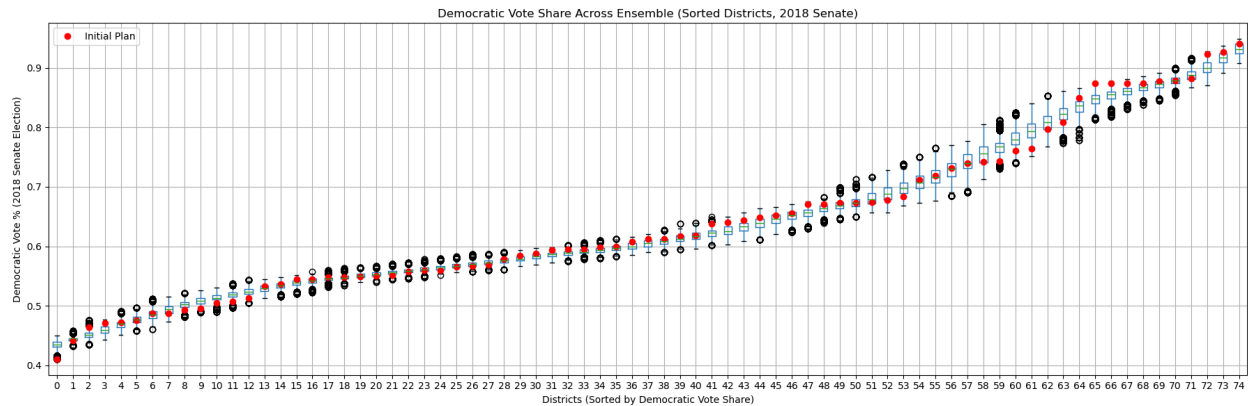
- **Interpretation:** The average of 6 majority-Latino districts suggests most plans produce around 6 such districts, reflecting Latino population distribution. A peaked histogram around 6 shows stability, meaning Latino-majority districts are geographically constrained. A wider or multi-peaked histogram would indicate more flexibility in drawing these districts.
- **Why It Matters:** This is crucial for assessing potential Latino representation under the Voting Rights Act. It shows whether the enacted plan provides typical, enhanced, or diminished opportunities for Latino-majority districts compared to the ensemble.

### 3. Democratic-Won Districts Histogram:

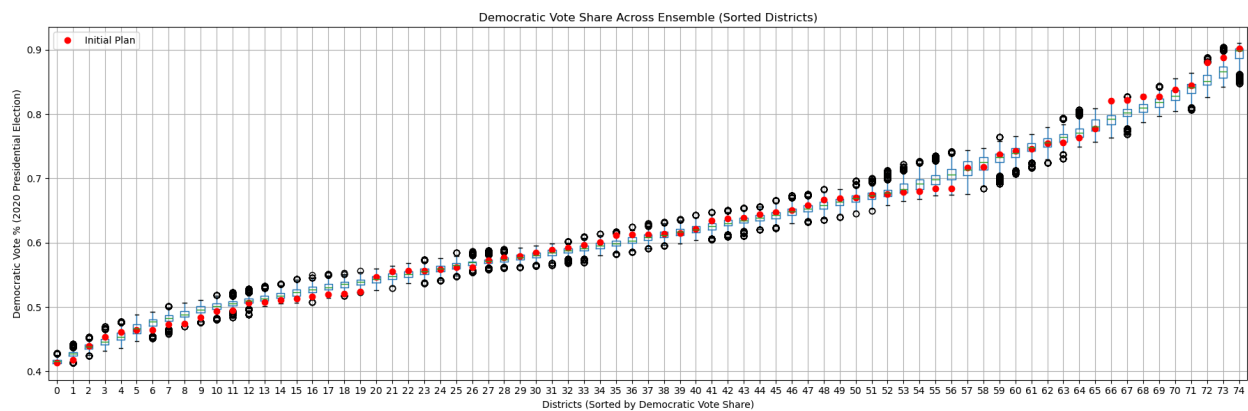
- **What It Shows:** Displays the frequency of plans with different numbers of Democratic-won districts based on the respective election data.
- **Interpretation (2018 Senate Data):** The number of Democratic wins in the initial plan (65) is slightly lower than that in most simulation plans (ensemble average 67), and there may be a slight Republican tilt (less favorable to Democrats).
- **Interpretation (2020 Presidential Data):** The initial plan (63 Dem wins) is on the left side of the simulation distribution (ensemble average 64), showing a slight deviation; but it is not seriously skewed.
- **Why It Matters:** This assesses partisan fairness. If the enacted plan's outcome is an outlier compared to the ensemble, it could suggest potential partisan gerrymandering. The results here indicate the enacted plan is slightly less Democratic-performing than the average neutrally drawn plan for both election datasets.

### 4. Efficiency Gap (EG) Distribution Histogram:

- **What It Shows:** Displays the frequency of different EG values across the ensemble.
- **Interpretation (Both 2018 & 2020 Data):** Suggests that the initial proposal is slightly less favorable to Republicans than the simulations. But both simulation plan results and initial plan results exceed 8%, they are about 10% - 13%.



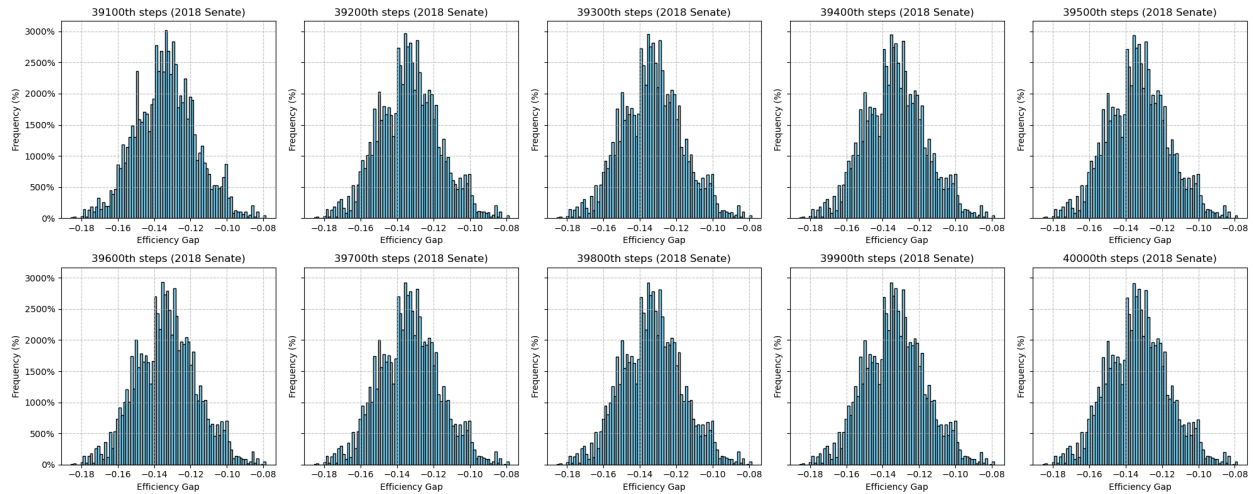
**Figure 7: 2018 Marginal Box Plot**



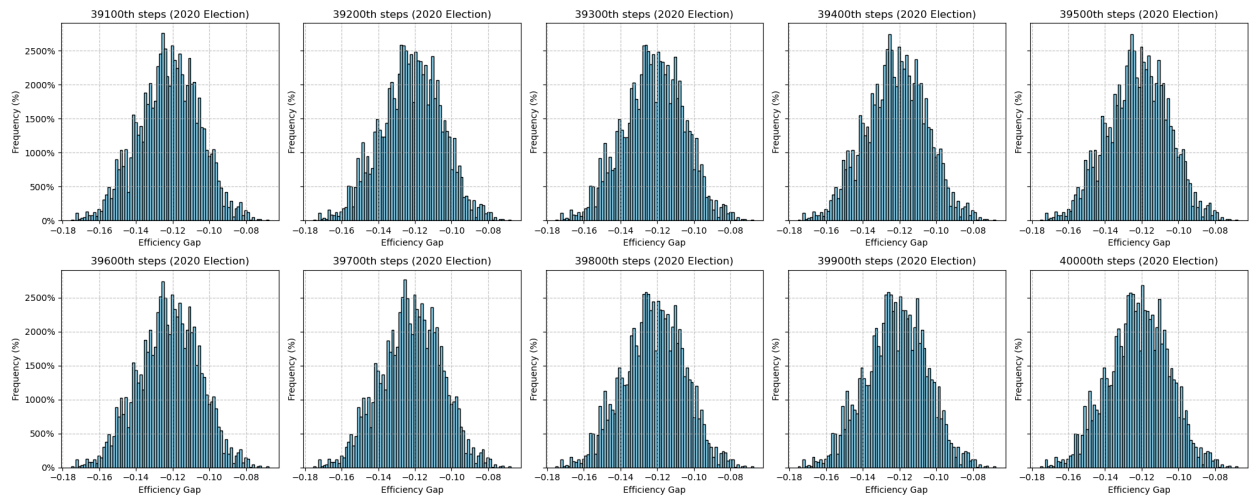
**Figure 8: 2020 Marginal Box Plot**

### Democratic Vote Share Box Plots (Sorted Districts):

- **What It Shows:** These plots display box plots for each of the 75 districts, sorted by their Democratic vote share in the enacted plan. Each box plot summarizes the distribution of Democratic vote percentages for that district across the ensemble. Red dots indicate the Democratic vote share for that district in the enacted plan.
- **Interpretation (Both 2018 & 2020 Data):** Both Box plots indicate that the initial plans for both elections generally fall within the ensemble distribution, but the 2018 plan shows slightly more variation away from neutral expectations compared to the 2020 plan.
- **Why It Matters:** This helps visualize if individual districts in the enacted plan are partisan outliers compared to what could be achieved under neutral redistricting.



**Figure 9: 2018 Convergence (Step 39,100 - 40,000)**



**Figure 10: 2020 Convergence (Step 39,100 - 40,000)**

### Efficiency Gap Convergence Histograms (from 40,000-step chains):

- **What It Shows:** A series of histograms showing the evolving distribution of the Efficiency Gap at different stages of the longer 40,000-step Markov chain (specifically for the final 1,000 steps shown in intervals of 100).
- **Interpretation (Both 2018 & 2020 Data):** The distributions of efficiency gap remain consistent in the final 1,000 steps for both elections, confirming that the Markov Chain has converged. This validates the reliability of the ensemble-based analysis.
- **Why It Matters:** These plots are crucial for ensuring that the MCMC simulation has run long enough to adequately explore the space of possible plans, meaning the resulting distributions of metrics like the EG are stable and reliable.



## 5. Discussion

The GerryChain analyses provide insights into Rhode Island's State House districting plan:

- The enacted plan is population-balanced. Its compactness, as measured by cut edges, appears poorer than the ensemble average for both election datasets.
- In terms of partisan outcomes, the enacted plan results in slightly fewer Democratic-won districts than the ensemble average for both the 2018 Senate and 2020 Presidential elections, suggesting it may be slightly less favorable to Democrats (or slightly more favorable to Republicans) than typical neutrally drawn plans.
- The Efficiency Gap for both the enacted plan and the ensemble averages indicates a Republican advantage (negative EG values around -10% to -13%).
- The number of majority-Latino districts (around 7 on average in the ensembles) appears relatively stable, likely reflecting the geographic distribution of the Latino population and limiting variability in creating such districts.

### Limitations noted include:

- The presence of a disconnected node (island) in the input geography, which might slightly skew results if it carries population or votes.
- The population deviation tolerance (30%) is relatively high and could allow for less equitable plans in terms of population equality than a stricter tolerance.
- The use of `accept=always_accept` for valid proposals means the ensemble samples a wide range of plans, some of which might be unrealistic in shape if not for additional (unapplied) compactness constraints.

## 6. Conclusion

The GerryChain analyses offer a clear, quantitative view of Rhode Island's redistricting possibilities based on the 2018 Senate and 2020 Presidential election data. The histograms and other visualizations provide a valuable baseline for evaluating the enacted State House plan's characteristics concerning compactness, potential for minority representation, and partisan fairness.

The findings suggest that while the enacted plan is population-balanced, it is less compact than typical neutrally drawn plans. Its partisan performance is slightly less Democratic than the ensemble averages for both election cycles analyzed. The Efficiency Gap indicates a Republican lean for both the enacted plan and the ensemble.

Future work could explore the impact of stricter population deviation constraints, incorporate compactness constraints into the ensemble generation, run longer

simulations for all metrics (not just EG convergence), or analyze additional metrics such as the number of competitive districts to further deepen the fairness analysis.

## **Short Burst Analysis of Rhode Island's Redistricting**

### **1. Introduction**

This section extends the previous GerryChain analysis by applying shortburst optimization to maximize the number of Latino-majority districts in Rhode Island's State House, addressing representation goals under the Voting Rights Act.

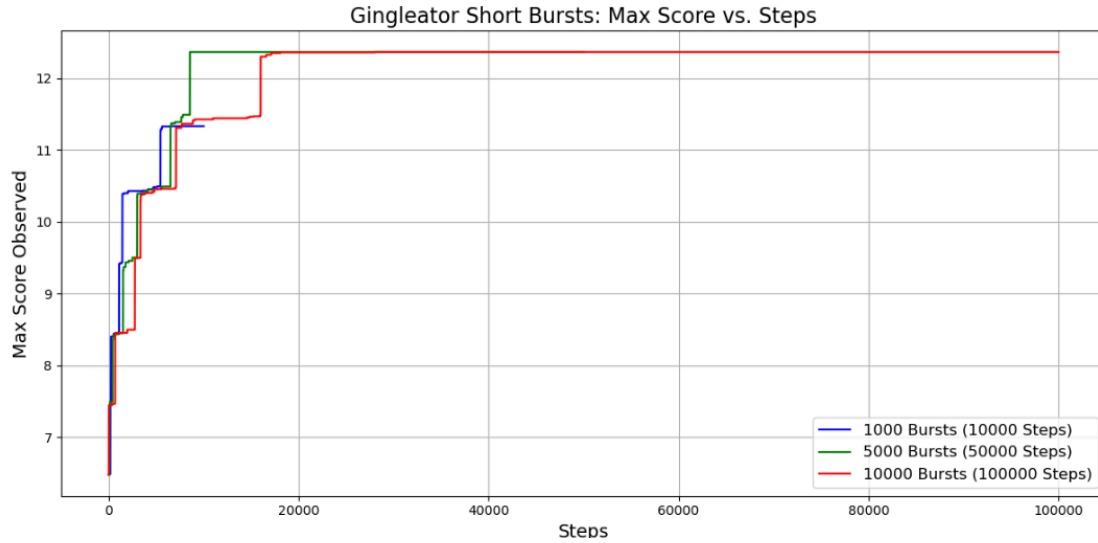
To maximize Latino representation in Rhode Island's 75 State House districts, we employed a shortburst optimization approach using GerryChain's Gingleator class, inspired by the `redist_shortburst` algorithm (Cannon et al., 2020). This method runs short MCMC simulations (bursts) to propose district plans, evaluates them using a scoring function, and restarts from the best plan to efficiently explore the solution space.

### **2. Methodology**

We configured Gingleator with the `reward_partial_dist` function, which scores plans by the number of Latino-majority districts (Hispanic population >50%) plus partial credit for the district with the highest Latino percentage below the threshold. Each burst comprised 10 MCMC steps using the ReCom proposal to merge and split adjacent districts, ensuring population balance within 28% of the ideal (10,544.85–18,718.60). We ran three simulations for 1,000, 2,000, and 10,000 bursts (10,000, 20,000, and 100,000 total steps), accepting all proposed plans to sample diverse configurations.

### **3. Results**

The shortburst algorithm converged to a local maximum of 12–13 Latino-majority districts within 5,000 bursts across all runs, as shown in the Figure. This exceeds the ensemble's average of 6 majority-Latino districts (Section 4), reflecting the potential to enhance Latino representation beyond typical plans.



**Figure 10: Gingleator Short Burst Result**

## 4. Discussion

The 12–13 Latino-majority districts support stronger compliance with the Voting Rights Act, ensuring greater opportunities for Latino representation without significantly altering the partisan balance (~64 Democratic-won districts, per Section 4). However, confirming this as the global maximum is infeasible due to the combinatorial complexity of redistricting and the heuristic nature of shortburst, which may miss better plans in unexplored regions (Cannon et al., 2020). Comparing results to the ensemble’s histogram (Section 4) shows 12–13 districts are at the upper end of feasible plans, suggesting a strong but not necessarily optimal outcome.

## 5. Future Work

Future analyses could increase steps per burst, enforce compactness constraints, and use Latino voting-age population to better reflect electoral influence. Addressing the disconnected node and tightening population tolerance would enhance results. Comparing our shortburst to the `redist_shortburst` algorithm or running multiple optimizations with varied initial plans could validate the local maximum’s robustness.